🐍 What is pip in Python?

pip stands for **"Pip Installs Packages"** (or sometimes "Pip Installs Python").
It's the **default package manager** for Python — used to **install, update, and manage** external libraries and dependencies that aren't part of Python's standard library.

🧰 Why do we use pip?

Python comes with many built-in modules (like math, os, datetime), but sometimes you'll need **external libraries** such as:

- requests → for HTTP requests
- pandas → for data analysis
- flask → for web development
- numpy → for numerical operations

You can install these using pip.

⚙️ Basic pip Commands

Command	Description
pip --version	Check pip version
pip install package_name	Install a package
pip uninstall package_name	Uninstall a package
pip list	List all installed packages
pip show package_name	Show info about a package
pip install --upgrade package_name	Upgrade a package
pip freeze	Show installed packages (for requirements.txt)

💡 Examples

1️⃣ Install a package

pip install requests

2️⃣ Check if installed

pip show requests

3️⃣ Uninstall a package

pip uninstall requests

4️⃣ Upgrade a package

pip install --upgrade requests

📦 Installing from a requirements.txt

When working on projects, you may see a file called requirements.txt that lists all dependencies.

Example:

flask==3.0.0

requests==2.31.0

pandas

To install everything:

pip install -r requirements.txt

👷 Where does pip install packages?

Pip installs packages into:

- The **global environment** (system-wide), or
- A **virtual environment** (recommended for projects)

🧱 Using pip in a Virtual Environment

It’s best practice to install packages in a **virtual environment** so that project dependencies don’t conflict.

```
# Create a virtual environment

python -m venv myenv


# Activate it

# On Windows:

myenv\Scripts\activate

# On Mac/Linux:

source myenv/bin/activate


# Now install packages

pip install flask
```